

Spec-First

Humans edit

Spec + Code

Spec maintained?

Optional — may be discarded

Testing requirement

Standard

Tooling maturity needed

Low

Best for

Prototypes, proof-of-concepts

Spec-Anchored

Humans edit

Spec + Code (both maintained)

Spec maintained?

Yes — synchronized with code

Testing requirement

Spec-aligned tests required

Tooling maturity needed

Medium

Best for

Long-lived production systems

Spec-as-Source

Humans edit

Spec only — code is generated

Spec maintained?

Yes — it IS the source of truth

Testing requirement

Full automated suite (unit, integration, e2e)

Tooling maturity needed

High

Best for

Mature teams with trusted generation tooling

Increasing maturity and automation